A Candidate for an Unconditional Computer-Assisted Proof of

the de Bruijn-Newman Upper Bound

Lambda <= 0.1787854

Status boundary

This is an unreviewed computer-assisted candidate for an unconditional proof. It is not an established theorem, has not been independently validated, and is not a proof of the Riemann hypothesis. The word candidate is load-bearing.

Field	Value
Prepared for	Adversarial mathematical and computational review
Prepared by	Jude Gomila
Primary repository	https://github.com/judegomila/dbn-lambda-01787854-candidate-audit
Evidence binding	PDF listed in repository SHA256SUMS; use the exact accompanying Git commit

The manuscript summarizes the proof architecture and review contract. The repository source files and published references remain authoritative.

Abstract and review status

This manuscript presents a private referee package for a proposed computer-assisted upper bound on the de Bruijn-Newman constant. The proposed argument instantiates Theorem 1.2 of D. H. J. Polymath with exact parameters, uses the published Platt-Trudgian finite verification of the Riemann hypothesis, and supplies new finite-window, all-N tail, and closed-barrier arguments with directed interval certificates.

```
X       = 6000000185827
t0      = 129/800 = 0.16125
y0^2    = 87677/2500000 = 0.0350708

t0 + y0^2/2 = 893927/5000000 = 0.1787854
```

If every cited theorem is applicable, every new analytic lemma is correct, and every implementation faithfully certifies its stated predicate, the assembled argument would establish Lambda <= 0.1787854 without assuming an unproved conjecture. Those antecedents have not yet been independently confirmed.

The package claims	The package does not claim
An exact candidate theorem and a complete internal dependency chain.	Independent acceptance of any new analytic reduction.
Directed interval certificates for every encoded finite, tail, and barrier gate.	A proof of the Riemann hypothesis or Lambda = 0.
Fresh replay paths and a fail-closed release seal.	Peer review, publication, novelty, or public priority.
A proposed unconditional logical form.	That green software output alone proves unconditionality.

Decision requested from the referee

Audit the published-theorem applications, handwritten analytic reductions, proof-to-code maps, and directed computations as separate layers. Report whether each of the three exact Polymath criterion hypotheses is established.

Contents

How to use this manuscript

Sections 1-9 state the proposed mathematics. Sections 10-12 describe evidence, replay, and review procedure. Sections 13-14 state the residual risks and disposition. The appendices bind headline values to repository paths.

1. Criterion and exact candidate row

Let H_t denote the heat-flow deformation of the Riemann xi function in the normalization used by the Polymath paper. The de Bruijn-Newman constant Lambda is characterized by the property that the zeros of H_t are all real precisely for t >= Lambda. The proposed proof uses the upper-bound criterion in Polymath Theorem 1.2 rather than attempting a new global characterization.

Item	Exact value	Role
Barrier abscissa X	6000000185827	Criterion site
Required zeta height	X/2	Hypothesis (i)
Time t0	129/800	Final time and barrier top
Height square y0^2	87677/2500000	Criterion geometry
Candidate bound	893927/5000000	t0 + y0^2/2
Final y ceiling	sqrt(271/400)	Hypothesis (ii)

Three hypotheses to be supplied

Criterion leg	Exact domain	Proposed supplier
(i) verified height	0 <= T <= X/2; sigma to the right of 1/2	Platt-Trudgian plus exact H0/xi sign map
(ii) final-time region	x >= X+sqrt(1-y0^2), y0 <= y <= sqrt(1-2t0)	Finite Triangle windows plus all-N tail
(iii) curved barrier	0 <= t <= t0 on the criterion barrier	Closed rectangular zero-free certificate

```
y0^2 + 2 t0 = 893927/2500000 < 1
y0^2 < 1 - 2 t0 = 271/400
0 < t0 < 1/2,  X >= 200
```

All parameter identities and domain containments are checked with exact rational arithmetic before the final assembly can reach its conclusion.

2. Proof architecture and dependency discipline

The proposed argument is deliberately split into cited inputs, new mathematics, numerical realizations, stored evidence, and a final logical weld. No numerical transcript is permitted to stand in for an unstated theorem.

Layer	Object	Failure meaning
Published input	Polymath Theorems 1.2 and 1.3; Platt-Trudgian Theorem 1	Candidate cannot be assembled as stated
New analytic layer	Native binding, Dini transfer, window freeze, tail, derivative and barrier lemmas	Mathematical gap even if every replay is green
Implementation	C/FLINT/Arb and Python interval programs	Proof-to-code correspondence failure
Evidence	Finite shards, tail logs, coefficient and prism records	Stored certificate is incomplete or inconsistent
Assembly	verify_assembly_1787854.py	One or more exact criterion legs is not supplied

Dependency flow

```
Platt-Trudgian verified height ---------------------------> (i)

finite rows -> native binding -> all-y transfer
            -> window freeze -> effective error --+
                                                   +-----> (ii)
all-N tail theorem -> 256/512-bit Arb certificate --+

stored coefficients -> uniform error -> derivatives
                    -> 883 closed prisms -> winding ------> (iii)

(i) + (ii) + (iii) + exact rational substitution
    -> candidate conclusion Lambda <= 0.1787854
```

Fail-closed rule

Malformed evidence, missing rows, nonconsecutive seams, interval overlap at a strict comparison, unexpected output, environment overrides, and manifest drift are fatal. A failed gate cannot be downgraded to a warning by the final assembly.

3. Verified-height hypothesis

Platt and Trudgian rigorously verify, using interval arithmetic, that every nontrivial zeta zero through height T_PT = 3000175332800 lies on the critical line. The candidate criterion needs information only through X/2.

```
T_PT = 3000175332800
X/2  = 6000000185827/2 = 3000000092913.5

T_PT - X/2 = 350479773/2 = 175239886.5 > 0
```

Normalization and sign map

A zero H_0(x+iy)=0 first maps to a zero of xi at (1-y+ix)/2. The xi functional equation followed by conjugation produces the representative (1+y+ix)/2 required by the criterion. The affine identities are checked over exact rationals by verifiers/verify_criterion_sign_map.py.

The cited finite verification handles positive ordinate. At T=0 and 0<sigma<1, the alternating eta series is strictly positive after grouping consecutive terms, while eta(sigma)=(1-2^(1-sigma)) zeta(sigma) has a strictly negative prefactor. Thus zeta(sigma)<0; at sigma=1, zeta has a pole. This closes the real-height endpoint of hypothesis (i).

Check	Result
x-to-height normalization	x = 2T
critical line	y = 0
required real part	(1+y0)/2 > 1/2
upper endpoint	s = 1 is a pole, not a zeta zero
height surplus	350479773/2

Cited theorem boundary

The repository does not re-prove the Platt-Trudgian computation. The referee must verify that its published theorem has exactly the strength and normalization consumed here.

4. Finite final-time corpus

At t=t0 the finite lane assigns one half-open x-window to each integer N from 690988 through 3840000. Fifteen compressed certificate shards contain exactly one successful row for every N, with no gap, duplicate, overlap, nonpositive row, or UNCERT record.

Family	Auxiliary primes	N range	Rows	Stored minimum
P11	2,3,5,7,11	690988..728999	38,012	0.000000791366
P7	2,3,5,7	729000..818999	90,000	0.000315112459
P5	2,3,5	819000..1027999	209,000	0.000305788807
P23	2,3	1028000..3840000	2,812,001	0.000309285478

```
number of rows = 3840000 - 690988 + 1 = 3149013
global stored T floor = 0.000000791366 at N = 690988
effective error Emax <= 0.000000233494905212337849

finite margin >= 0.000000557871094787 > 0
```

Independent finite checks

A fresh producer run regenerated and compared every canonical row. Thirty direct 256-bit singleton evaluations cover the weakest row, both sides of each prime-family transition, both sides of every compressed-shard joint, and the closed finite-tail overlap at N=3840000. The weakest direct row reproduces the stored floor.

Tight margin

The finite margin is rigorous but small. A reviewer should prioritize directed rounding, the native normalization units, the all-height transfer, and every source branch at the weakest P11 row.

5. Native binding, height transfer, and window freeze

The stored Triangle floor must be connected to the normalized function f_t in Polymath Theorem 1.3 before its sign can be used. NATIVE_BINDING.md supplies exact real-coefficient Dirichlet convolutions for both Riemann-Siegel sums.

```
E A        = sum B_(N,n) n^(-s_*)
conj(E) C0 = sum A_(N,n) n^(-conj(s_*))

|f_t| >=
  [1 - g_N - sum_(n>=2)(|B_(N,n)|+g_N|A_(N,n)|)n^(-sigma_N)]
  / M_N - g_N C_N
```

Positivity supplies the sign required when replacing |E| by its upper bound M_N and also rules out E=0. No second Euler-factor conversion is applied. The resulting T_N is in the same normalized units as the additive Theorem 1.3 error.

Conservative effective-error weld

The target proof does not use the 10.44 constant in displayed equation (24). Starting from Proposition 6.6(vi), it applies 1+u <= exp(u), enlarges x-8.52 to x-12 in the denominator, and combines the exact constants 3.58+6.92=10.50. ERROR_CONSTANT_WELD.md records the derivation and a fail-closed checker verifies all six numerical consumers.

Authoritative Arb error budget

The uniform effective-error budget Emax and the finite margin are certified authoritatively by the standalone FLINT/Arb program verifiers/verify_prop410_arb.c, run at 256 and 512 bits in the pinned container (transcripts logs/prop410_arb_256.log and logs/prop410_arb_512.log, strictly parsed as assembly prerequisite P17, with a twenty-mutation fail-closed suite). Every decisive inequality subtracts the exact rational bound and requires the whole resulting ball on the strict side. The mpmath.iv computation of the same budget, and its copy under independent/prop410/, are same-backend corroboration only.

All-y transfer

The direct upper-Dini proof retains the signed composite-divisor cancellation, exhausts 243, 81, 27, and 9 active sign cells for the four prime families, and gives worst ratio <= 0.99999860767275095. Separate interval arguments prove the normalizer and kappa correction nonincreasing.

Corrected x-window minorant

```
Sigma(x,y) =
  (1+y)/2 + (t0/4) log(x/(4 pi))
  - t0/(2 x^2) [1-3y+4y(1+y)/x^2]_+

G(x,y) <= G_N(y)
K(x,y) <= K_N(y)
Sigma(x,y) >= Sigma_N(y)
```

The plus sign before the logarithmic term is essential. The window-freeze theorem proves Sigma increasing in x, including across the positive-part kink. Exact pi and square-root bounds place X+sqrt(1-y0^2) strictly inside the N=690988 window.

6. Infinite all-N tail theorem

The finite lane ends only after the complete N=3840000 window; the tail begins at the same closed cutoff. TAIL_LEMMA.md gives a quantified theorem for every N at or above this cutoff, the complete required height interval, and a closed time box containing t0. It is not a sampled extrapolation.

Quantity	Certified directed result
Closed cutoff	N_* = 3840000
Convolution head	M = 153814
Error head	M_err = 3000
Contraction	D < 0.999721
Mollifier cap	M_max < 1.608290
Normalized flow	> 0.0001735326089372
Effective error	< 0.000000011671604
Post-error margin	> 0.0001735209373337

Reduction to the cutoff

An exact finite Dirichlet convolution is partitioned into disjoint routed classes. Endpoint-cap lemmas and checked derivative signs reduce all N to the cutoff. Moving floors and N/(d+1) terms are handled explicitly. The y transfer covers y0 through sqrt(1-2t0), and the complete t box is interval evaluated without assuming monotonicity in t.

```
|M_lambda(s_*) f_t - 1| <= D < 1

(1-D)/M_max - (e_A + e_B + e_C0)
    > 0.0001735209373337 > 0
```

One standalone FLINT/Arb implementation run at 256 and 512 bits is the primary certificate and is independently parsed. Python interval runs at 160 and 256 bits provide a separate implementation and lineage check.

7. Closed barrier: approximation and coefficient data

The proposed barrier certificate proves zero-freeness on a closed rectangle that contains the complete curved barrier required by Polymath Theorem 1.2.

```
R = [X, X+1] + i [1809/10000, 1]
0 <= t <= 129/800

H_t(z) != 0  for z in R
```

Barrier fact	Certified value
Floor-square margin	y0^2 - 0.1809^2 = 234599/100000000 > 0
Riemann-Siegel index	N = 690988 throughout the box
Total error using Proposition 6.6(vi) corollary	< 0.000356523011600040
Common approximation allowance	1/800 = 0.00125
Stored complex matrix	62 x 62
Real components checked	7688/7688 contained
Omitted Taylor tail	< 1.954234593244762 x 10^-22

The t=0 endpoint

Polymath Theorem 1.3 is stated for positive t. The barrier note extends the inequality to t=0 on this fixed compact rectangle: dominated convergence gives continuity of H_t, the fixed Riemann-Siegel index makes f_t a finite continuous sum, the conservative majorant is continuous, and B_t and its reciprocal are continuous and nonzero.

Coefficient provenance

Each stored 20-decimal component is restored as a ball with radius 10^-20 max(1,|component|). A separate Arb producer regenerates every component inside that allowance; the factorial Taylor remainder is propagated into every barrier value.

8. Derivative bounds, closed prisms, and winding

At each exact time seam the producer encloses f_t on a cyclic boundary mesh. Uniform spatial and time derivative majorants control the complete rectangle and the complete proposed time prism. The acceptance gate is one strict interval inequality.

Although Polymath Lemma 8.4 is stated for t>0, the fixed-N finite sums and their displayed derivatives are C1 on the closed box, with denominators and logarithm branches certified safe. Applying the lemma for t>=epsilon and then taking epsilon down to zero therefore supplies the same derivative majorants at the initial seam.

```
M_i >
  D_(z,i) / [2 (num-1)]
  + D_(t,i) (t_(i+1)-t_i)
  + 0.00125
```

The factor 1/2 follows from the nearer-endpoint convex-disk homotopy on each half-subedge. D_t is recomputed on the whole proposed prism, rather than sampled at its left endpoint. Every endpoint ball must exclude zero, every argument increment must lie strictly inside (-pi,pi), and the winding enclosure must lie strictly inside (-1/4,1/4).

Recorded barrier result	Value
Closed time prisms	883 consecutive
Initial time	exactly 0
Final endpoint	contains 129/800
Minimum recomputed prism margin	> 0.519849894613872543
Aggregate winding enclosure	inside [-8.95,8.95] x 10^-13
Linux replay	GCC / FLINT 3.0.1; 54 parser checks
macOS replay	Clang / FLINT 3.6.0; 54 parser checks

The zero-avoiding spatial, time, and approximation homotopies carry winding zero from the stored polygon to H_t/B_t on the true boundary. Because B_t is nonzero, the argument principle gives no H_t zeros inside the rectangle throughout each prism.

The last stored dyadic prism may extend slightly beyond the exact t0. The theorem restricts that already-certified prism to its intersection with 0<=t<=129/800. Restriction only decreases the displacement allowance, and the uniform approximation estimate is used only through exact t0.

9. Final theorem weld

The final assembly executes every stored prerequisite before checking the exact domains and reaching the canonical Polymath criterion weld.

Hypothesis	Proposed evidence	Exact coverage result
(i)	Published verified height plus sign map	X/2 is below T_PT by 350479773/2
(ii), finite	3,149,013 rows plus binding and transfer	[x_*, x_3840001)
(ii), tail	All-N tail theorem and Arb certificate	[x_3840000, infinity)
(iii)	883-prism closed rectangle	Contains the entire curved barrier for 0 <= t <= t0

```
(i) + (ii) + (iii)
    => Lambda <= t0 + y0^2/2

       = 129/800 + (87677/2500000)/2
       = 893927/5000000
       = 0.1787854
```

Candidate conclusion

The repository contains an internally complete proposed proof chain. This conclusion remains a candidate until external mathematicians validate the published-theorem applications, new lemmas, proof-to-code correspondence, and interval computations.

No conjectural premise

The finite Platt-Trudgian result is a published computational theorem, not an assumption of the full Riemann hypothesis. Thus the intended logical form is unconditional. Calling the candidate unconditional does not prejudge whether the new argument is correct.

10. Evidence hierarchy and integrity seal

A computer-assisted proof must distinguish the proposition, its reduction, the implementation, the numerical evidence, the parser, and the release binding. Agreement in later layers cannot repair an error in an earlier one.

Evidence level	Question answered	Principal artifact
1. Statement	What mathematical fact is needed?	Proof notes
2. Reduction	Why does a finite check suffice?	New lemmas
3. Implementation	Does code encode that reduction?	C/Python source
4. Directed output	Did strict inequalities hold?	Logs/certificates
5. Parser/assembly	Was all evidence consumed?	Fail-closed verifiers
6. Release binding	Which exact bytes were reviewed?	SHA256SUMS + Git

Seal behavior

scripts/seal.py inventories every stable regular file except the manifest itself and explicitly transient roots. An unlisted file, missing file, digest mismatch, symlink, special file, malformed manifest entry, or forbidden build product is fatal. The manifest must be regenerated only after the final PDF and every source or documentation edit are complete.

Adversarial checks already represented

- finite row gaps, duplicates, overlaps, nonpositive floors, and UNCERT records;
- direct singleton checks across every family and shard seam;
- tail precision agreement, malformed balls, wrong domains, and missing gates;
- barrier seam continuity, terminal coverage, winding, and coefficient provenance;
- ambient Python and barrier override variables that could weaken checking;
- sanitizer, compiler-warning, and negative-input paths for critical producers.

Interpretation

A green seal proves artifact identity. A green replay proves the encoded predicates. Neither result, by itself, validates the mathematical reductions.

11. Reproduction protocol

Level A: stored fail-closed verification

```
python3 -m venv .venv
. .venv/bin/activate
python3 -m pip install --require-hashes -r requirements.txt
./verify.sh
```

This checks the release seal, historical provenance subset, all stored finite rows, native binding, window freeze, tail, both barrier transcripts, sign map, and exact final assembly.

Level B: portable clean container

```
docker build --platform linux/amd64 -t dbn-lambda-01787854-review .
mkdir -p replay/container-review
docker run --rm --platform linux/amd64 --network none --read-only \
  --cap-drop ALL --security-opt no-new-privileges \
  --user "$(id -u):$(id -g)" \
  --tmpfs /tmp:rw,exec,nosuid,size=4g \
  -e REVIEW_OUTPUT=/review-output/evidence \
  -v "$PWD:/work:ro" \
  -v "$PWD/replay/container-review:/review-output" \
  -w /work dbn-lambda-01787854-review
```

Levels C-E: fresh producers

```
./scripts/run_barrier_replay.sh replay/barrier
./scripts/run_tail_arb.sh replay/tail_arb

review_image_id=$(docker image inspect --format '{{.Id}}' \
  dbn-lambda-01787854-review)
IMAGE=dbn-lambda-01787854-review \
EXPECTED_IMAGE_ID="$review_image_id" \
  ./scripts/run_full_sweep.sh replay/full_sweep
```

The full sweep regenerates all 3,149,013 finite rows. The barrier replay regenerates all 7,688 coefficient components, Taylor and uniform-error bounds, and all 883 prisms. The tail replay builds and runs the standalone Arb implementation at 256 and 512 bits.

> **Resource orientation**
>
> The sealed GitHub workflow required roughly one hour for each of its full-review and all-row lanes. Runtime is hardware-dependent; review conclusions must be attached to the exact commit and transcripts, not to timing values.

12. Referee protocol

The highest-value review is conceptual. Replaying code first can confirm package usability, but the proof decision should follow the dependency order below.

Priority	Referee task	Decision to record
1	Compare the exact parameter row and all three domains with Polymath Theorem 1.2.	Correct theorem transcription and weld?
2	Audit the closed-barrier derivative, interpolation, winding, B_t, and t=0 arguments.	Complete zero-free homotopy?
3	Check coefficient rounding, Taylor remainder, and Theorem 1.3 error formulas.	Valid common 0.00125 allowance?
4	Check native Triangle convolution, signs, normalization, and source map.	Stored floor bounds normalized |f_t|?
5	Check Dini transfer, window freeze, endpoint convention, and finite-tail overlap.	Complete finite domain?
6	Check all-N tail partition, caps, monotonicity, and effective error.	Complete infinite domain?
7	Reproduce decisive computations in a different build environment.	Independent numerical agreement?

Required final report

- state separately whether hypotheses (i), (ii), and (iii) are established;
- identify any conjectural premise, circular check, or shared conceptual dependency;
- classify each finding as documentary, local, bound-invalidating, or fatal to the method;
- record exact file, line, commit, toolchain, and transcript references;
- state whether the package may advance beyond unreviewed proof-candidate status.

13. Risk register and trust boundary

Risk	Current mitigation	Residual external question
Published theorem mismatch	Versioned PDFs, theorem maps, exact sign-map checker	Does the cited statement exactly imply the consumed form?
Finite normalization/sign error	Symbolic binding, source probes, direct rows	Is every inequality direction and unit correct?
All-height or all-N gap	Dini cells, cap derivatives, exact endpoint coverage	Are all monotonicity and floor arguments valid?
Barrier interpolation gap	Closed-prism derivative bounds and convex-disk proof	Does the homotopy cover every boundary/time point?
Coefficient truncation	All 7,688 components regenerated inside restored balls	Does the generator implement the intended formula?
Common implementation error	Multiple precisions, two platforms, Python/Arb paths	Are implementations independent enough?
Artifact drift	Fail-closed SHA-256 stable-tree seal	Was the exact reviewed commit preserved?

Most fragile numerical interfaces

The finite margin is approximately 5.58 x 10^-7, the worst Dini ratio is close to one, and the tail contraction is close to one. Each is still strictly certified, but all deserve independent directed recomputation.

What remains outside the machine seal

The release seal cannot establish that the handwritten lemmas are true, that the code realizes them faithfully, that independent paths avoid a shared conceptual mistake, or that the work is novel and publishable. Those are precisely the external review boundary.

14. Conclusion

The repository presents a coherent candidate theorem with exact parameters, three proposed criterion suppliers, fail-closed directed certificates, full finite regeneration, separate FLINT/Arb and Python tail implementations, dual-platform barrier transcripts, and a stable release seal.

```
PROPOSED RESULT

Lambda <= 893927/5000000 = 0.1787854
```

The proposed result remains an unreviewed candidate, not an established theorem. Its mathematical and computational interfaces require independent adversarial review.

Release disposition

Suitable for private adversarial review as a candidate for an unconditional computer-assisted proof. Not suitable for a claim of theorem acceptance until independent mathematical and computational reports are complete.

Recommended next actions

- give the reviewer access to this private repository at one exact review commit;
- freeze that review commit and retain all GitHub replay artifacts;
- obtain a line-by-line theorem report and a separate clean-environment computation report;
- repair every finding and rerun the complete seal and producer suite;
- only then prepare a public immutable archive, preprint, and journal submission.

References

[1] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann xi function, and a new upper bound for the de Bruijn-Newman constant*, Research in the Mathematical Sciences 6, 31 (2019), arXiv:1904.12438v2. Repository copy: references/polymath-1904.12438v2.pdf.

[2] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to 3 x 10^12*, Bulletin of the London Mathematical Society 53 (2021), 792-797, DOI 10.1112/blms.12460, arXiv:2004.09765v1. Repository copy: references/platt-trudgian-2004.09765v1.pdf.

[3] Mosaic Intelligence, *A certified unconditional upper bound Lambda <= 0.1875 for the de Bruijn-Newman constant*, DOI 10.5281/zenodo.21175533. Repository copy: references/dbn21a-main.pdf.

[4] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Transactions on Computers 66 (2017), 1281-1292, DOI 10.1109/TC.2017.2690633.

[5] The FLINT team, *FLINT: Fast Library for Number Theory*. flintlib.org. The primary replay uses FLINT 3.0.1; the macOS cross-toolchain transcript uses FLINT 3.6.0. See ENVIRONMENT.txt and Dockerfile.

Authoritative-source rule

Transcribed theorem maps and repository notes are review aids. The versioned published papers remain authoritative for cited results.

Appendix A. Exact numerical ledger

Obligation	Headline directed result	Primary evidence
Verified height	surplus 350479773/2	PROOF_NOTE.md; criterion assembly
Finite corpus	3,149,013 rows; min 0.000000791366	15 gzip shards; verify_finite_and_binding.py
Finite error	Emax <= 0.000000233494905212337849	logs/prop410_arb_256.log and _512.log (Arb, authoritative); logs/finite_and_binding.log
Finite margin	>= 0.000000557871094787	native binding and effective-error weld
Dini transfer	ratio <= 0.99999860767275095	180/256-bit direct Arb checks
Tail contraction	D < 0.999721	256/512-bit FLINT/Arb
Tail margin	> 0.0001735209373337	logs/tail_arb_256.log; tail_arb_512.log
Barrier approximation	error < 0.000356523011600040	barrier/certificates/uniform_error_256.log
Coefficient matrix	7688/7688 components contained	storedsum_provenance.log
Taylor remainder	< 1.954234593244762 x 10^-22	storedsum_taylor_tail.log
Closed barrier	883 prisms; margin > 0.519849894613872543	two barrier_target_closed transcripts
Final assembly	40 fail-closed assembly gates	logs/assembly_1787854.log

Rounded values in this appendix are orientation summaries. The directed endpoints in the stored logs and source predicates control every strict comparison.

Appendix B. Repository map

Path	Review purpose
PROOF_NOTE.md	End-to-end theorem chain and conclusion
CANDIDATE_PARAMETERS.md	Exact row, boxes, and margins
OPEN_REVIEW_QUESTIONS.md	Load-bearing referee decisions
NATIVE_BINDING.md	Triangle-to-normalized-function theorem
ERROR_CONSTANT_WELD.md	Conservative 10.50 Proposition 6.6(vi) reduction
WINDOW_FREEZE_THEOREM.md	Uniform x directions and endpoints
TAIL_LEMMA.md	All-N, all-y infinite-lane theorem
DERIVATIVE_BOX_LEMMA.md	Uniform barrier derivatives
BARRIER_CERTIFICATE.md	Closed-prism and winding theorem
certificates/	Complete compressed finite corpus
barrier/	Coefficient, error, source, and prism evidence
verifiers/	Fail-closed parsers and independent checkers
scripts/	Container and producer replay entry points
references/	Versioned authoritative paper copies
ADVERSARIAL_REVIEW_PROTOCOL.md	Dependency blueprint and falsification protocol
SECURITY.md	Private reporting and execution policy
CONTAINER_IMAGE.md	Pinned environment and exact verified image binding
SHA256SUMS	Stable-tree release binding

Paper regeneration

```
python3 -m pip install --require-hashes -r paper/requirements.txt
python3 paper/generate_paper.py
mkdir -p tmp/pdfs/rendered
pdftoppm -png -r 150 \
  output/pdf/dbn_lambda_01787854_candidate_audit.pdf \
  tmp/pdfs/rendered/page
pdfinfo output/pdf/dbn_lambda_01787854_candidate_audit.pdf
```

The generator uses ReportLab invariant mode and PDF standard fonts. The committed PDF must be rendered page by page and visually inspected before the final SHA-256 seal is written. Invariant metadata timestamps are synthetic; the visible date and accompanying Git commit identify the review snapshot.

End of referee manuscript

Please attach findings to the exact Git commit and distinguish artifact integrity, numerical replay, mathematical validity, and publication status.